

Automated Content Generation for Curriculum Aligned Education

A Web-Based Bilingual Learning Platform for Bangladeshi Students

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What Admin Does

1

Select Subject

Choose from NCTB curriculum subjects

2

Select Chapter

Pick specific chapter within subject

3

Define Topic

Specify exact topic to be covered

4

Feed NCTB Content

Upload textbook guidelines & constraints into RAG vector store (Qdrant)

RAG Knowledge Base

How Curriculum Accuracy is Enforced

STORE

NCTB textbook content chunked & embedded into Qdrant vector database

INDEX

Curriculum rules indexed: grade limits, topic scope, allowed question types

RETRIEVE

At generation time, relevant constraints fetched with semantic search

GROUND

LLM grounded in real curriculum — hallucination prevented

Teacher Worksheet Request



→ Request payload sent to Agentic Pipeline

Subject + Chapter + Topic

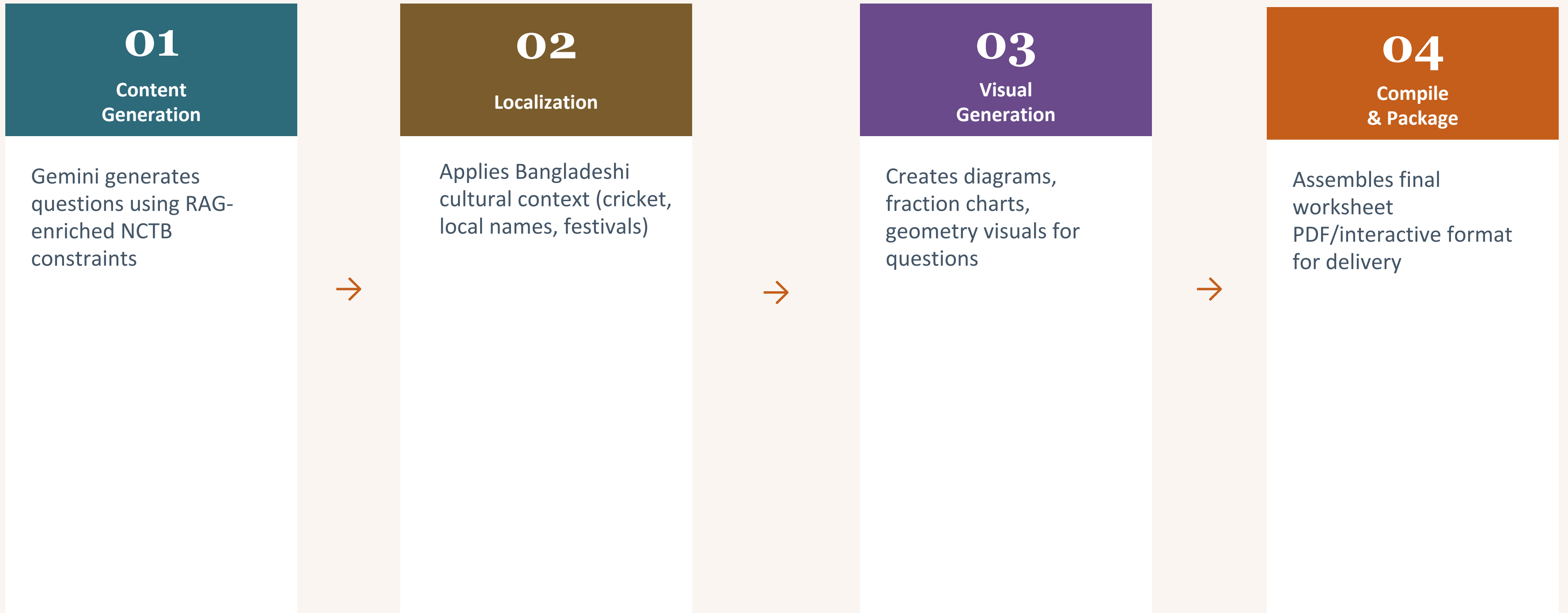
Sample worksheet (optional)

Class level

Language preference (EN/BN)

Difficulty setting

Sequential Agentic Pipeline



Each stage runs autonomously and passes output to the next agent — fully automated, no manual intervention needed

Generated Worksheet Preview

Q1. Addition

[Grade 3 · Math · Medium]

Q2. Word problem with diagram

[Grade 4 · Math · Medium]

Q3. Missing digit diamond puzzle,Solve triangle puzzle

[Grade 3 · Math · Medium]

Q4. Number and Count

[Grade 3 · Math · Medium]

Refinement Options

Teachers can modify before downloading:

Change Difficulty

Adjust Easy / Medium / Hard per question

Add New Problem

Insert AI-generated question on same topic

Add Visuals

Generate diagram or chart for any question

Remove Problem

Delete any question that doesn't fit

Edit Problem

Manually tweak question text or numbers

Regenerate All

Restart pipeline with different parameters

Download Final Worksheet (PDF / Interactive Web)

Static Worksheets

- Downloadable & printable PDF
- Suitable for offline learning
- Ideal for revision & assessments
- Supports Bengali & English

Visual / Diagram-Based

- Fraction pie & bar charts
- Geometry shape problems
- Coordinate grid worksheets
- AI-generated illustrations

Puzzle & Brain Teasers

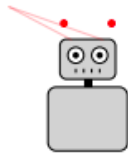
- Magic triangle problems
- Missing digit challenges
- Number pattern puzzles
- Higher-order thinking tasks



Math Fun: Addition

Class 3 | Topic: Addition of Two-Digit Numbers

Name: _____ Date: _____



Instructions: Solve the following addition problems by finding the missing digit.

1. Rahim bought 37 green mangoes from the market and his friend Karim gave him 2♦ more. If Rahim now has 62 mangoes in total, find the missing digit (♦):

$$\begin{array}{r} 37 \\ + 2\ \diamond \\ \hline 62 \end{array}$$

2. Fatima has ♦8 Taka in her piggy bank and her mother gave her 35 Taka more for her school tiffin. If she now has 73 Taka in total, find the missing digit (♦):

$$\begin{array}{r} \diamond 8 \\ + 35 \\ \hline 73 \end{array}$$

Mathematics Worksheet

Class: Class 3 | Subject: Mathematics | Topic: Addition of Two-Digit Numbers | Difficulty: Hard



Name: _____

Date: _____

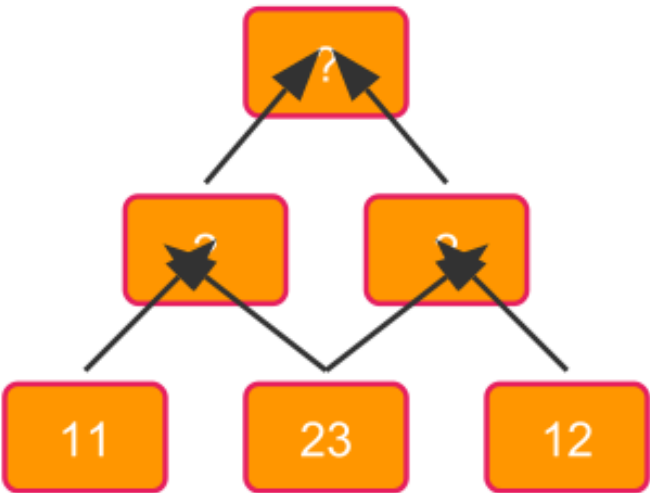
Solve the following addition problems.

1. Fill in the missing digits in the vertical addition problem:

$$\begin{array}{r} \boxed{?}5 \\ + 3\boxed{?} \\ \hline 87 \end{array}$$

Answer: _____

2. Solve this Number Pyramid puzzle. The rule is: Each block equals the sum of the two blocks directly below it. Fill in all the empty blocks.



Answer: _____

Name:
Date:



Addition Fun!

Class: Class 3 | Subject: Mathematics | Topic: Addition of Three-Digit Numbers

Instructions: Read each problem carefully and find the correct answer. Show your work!

Problem 1:

Fatima's school library received a special donation for Pohela Boishakh: 215 storybooks, 142 science books, and 301 art books. How many books were donated to the library in total?



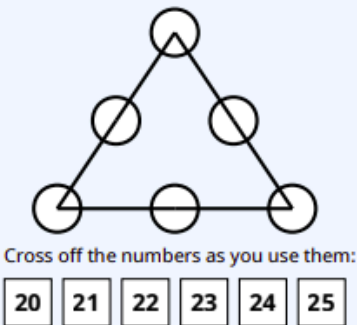
Answer:

Problem 2:

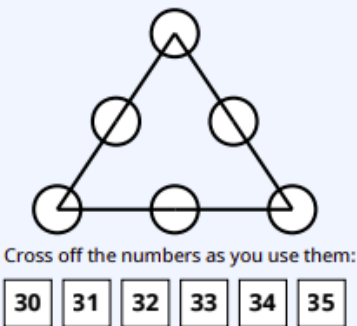
Fill in BOTH missing digits to make the addition correct: $4_3 + 2_5 = 678$

Answer:

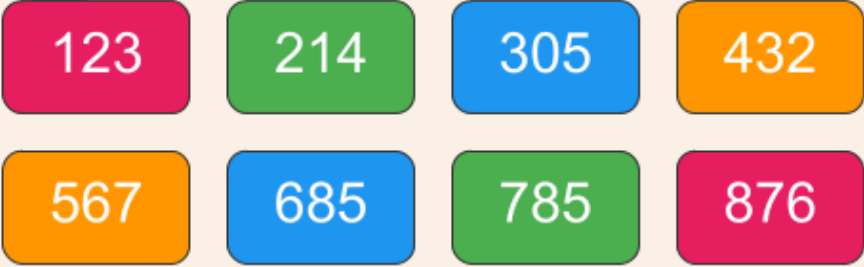
2. Rahim is helping his grandmother count out pieces of pitha for the Eid party. Help him place the numbers 20, 21, 22, 23, 24, 25 in the circles of the triangle so that each row (side) adds up to 66. Remember to cross off the numbers as you use them:



3. Fatima is organizing her school tiffin tokens. Help her place the numbers 30, 31, 32, 33, 34, 35 in the circles of the triangle so that each row (side) adds up to 96. Remember to cross off the numbers as you use them:



Can you find all pairs that add to 999?



Answer:

$$\begin{array}{r} \triangle 1 \\ + 3 \square \\ \hline 75 \end{array}$$

Now ,let's move on to the prototype:

Link:

Thank You

Questions & Discussion

- Admin feeds NCTB curriculum into RAG knowledge base
- Teacher selects subject/chapter/topic & triggers generation
- 4-stage agentic pipeline ensures accuracy & cultural relevance
- Interactive refinement before final download